

Healthcare Queue & Access App — Core Functionalities

● Priority 1 — Essential MVP

1. Healthcare Facility Map

- Show nearby clinics, hospitals and pharmacies.
- User selects a facility to view its page.
- Calculate approximate travel distance/time.

2. Current Queue Status

- Users submit the current queue/congestion level.
- Aggregate recent reports into:
 - ● Low
 - ● Moderate
 - ● High
- Show when the status was last updated.

3. Medication Reminder

- Users can input their medication consumption guidelines
- Get a reminder via whatsapp

4. Facility-Specific Forum

- Every facility gets its own community feed.
- Users post real-time updates.
- Posts can be upvoted/confirmed.
- Sort by newest/relevant.

5. Panic Button

- So facilities with emergency ambulances will have a button that immediately loads the number into the phone's dialer to call. To solve the issue of people not having emergency contacts of all facilities around them.
- Anywhere you go, you can always go to the app and click facilities with ambulances and just click and the number loads into your dialer.

6. Community Group Exercise Planner

- You can organize community exercise sessions on the app and after a threshold of people who want to join is reached then it is allowed to happen.

● Priority 2 — Trust & Intelligence

5. Location-Verified Reports

This is one of the features I'd definitely explore.

When someone submits a queue report, the app can check whether their phone is **near the selected facility**.

For example:

User submits: "Queue is very long"
GPS at submission: ~80 m from Clinic A
→ **Location verified ✓**

You don't need to reveal their location publicly.

You could also attach metadata such as:

- Distance from facility
- Timestamp
- How long they've been there
- Whether they are still nearby
- Number of other users currently detected nearby

This lets you distinguish between:

Verified current report

vs.

Unverified community report

Importantly, don't continuously track exact GPS unnecessarily. You can use **geofencing/proximity checks and coarse location**, retaining only what is needed.

6. Community Reliability

Use location + timestamps + confirmations/upvotes to establish report confidence.

For example:

● High congestion
8 verified reports · 5 min ago

rather than simply:

● High congestion
Someone reported this.

7. AI Forum Moderation

AI automatically:

- Detects spam/abuse.

- Categorises posts.
- Detects duplicate reports.
- Flags potentially important updates.
- Extracts useful queue information from natural language.

Example:

"Been here since 9, registration hasn't moved 🤔"

AI extracts:

Arrival: 09:00
Service: Registration
Reported delay: Significant

● Priority 3 — Advanced Data/ML

8. Predictive Waiting-Time Model

Once you have enough historical data, use:

Historical waiting times + time/day + current reports + verified presence + congestion

to estimate:

Typical waiting time: 1h 20m

Based on recent and historical data.

This is where the application starts becoming a genuine **data/ML product**.

9. Real-Time Congestion Detection

Use aggregated location metadata to estimate how many users are currently around a facility.

For example:

👤 ~24 active users nearby

● High congestion

You'd need to be careful here because **people near a clinic aren't necessarily patients in the queue**. Treat it as one signal rather than ground truth.

10. Anomaly Detection / AI Agent

An agent can continuously analyse incoming data and detect unusual situations:

Typical Tuesday: 40–70 min

Current reports: 150–180 min

12 verified users nearby

→ **Unusual congestion detected**

The system can then update the facility's status or flag it for review.

The Overall Architecture

Location data

→ verifies/weights user reports

Community reports

→ provide real-time information

Waiting-time data # Health & Community App — Feature Specification

1. Project Overview

The app is a **crowdsourced healthcare-access intelligence platform** designed to help South African communities understand what is happening at nearby healthcare facilities.

The platform focuses on:

- Finding nearby healthcare facilities.
- Understanding current queue and congestion conditions.
- Sharing facility-specific updates.
- Helping users manage basic health-related routines.
- Connecting communities through health and wellness activities.
- Making healthcare information accessible across multiple South African languages.

The app is **not intended to provide medical diagnoses or personalised medical advice**.

The core concept is:

**Find healthcare → understand what is happening → communicate with the community →
manage your health → participate in community health.**

2. Core Features

2.1 Facility Map & Information

Users can discover nearby:

- Clinics
- Hospitals
- Pharmacies
- Other relevant healthcare facilities

Each facility can have a dedicated information page containing:

- Address
- Contact details
- Opening hours
- Services offered
- Approximate distance

- Estimated travel time
- Emergency contact information where applicable

Mapping services such as Google Maps or another suitable mapping provider can be used for location and travel information.

2.2 Live Queue Status

Users can submit information about the current situation at a facility.

Possible report information includes:

- Approximate queue size
- Number of people waiting
- Estimated waiting time
- Type of queue/service
- Timestamp
- Optional description
- Optional location verification

Recent reports can be aggregated into a simple status:

-  Low
-  Moderate
-  High

The facility page can show:

 High congestion Approximately 40–60 people reported 9 recent reports Last updated 4 minutes ago

Older reports should gradually lose relevance so that the displayed status reflects recent conditions.

Queue size and waiting time should remain separate signals because a large queue does not necessarily mean a long waiting time.

2.3 Facility-Specific Forum

Each facility has its own community forum.

Users can post:

- Queue updates
- Facility/service updates
- Questions
- Medicine availability reports
- General discussions

Examples:

"Are blood pressure medicines available today?"

"Are they still accepting walk-ins?"

"The pharmacy queue is very long."

"The X-ray service is currently unavailable."

Other users can:

- Upvote posts
- Confirm information
- Reply
- Report inappropriate content

Posts can later be automatically classified using AI/NLP.

2.4 Emergency Call Button

For facilities with verified emergency/ambulance numbers, the facility page can provide an:

Emergency Call

button.

The button should:

1. Display the verified facility emergency number.
2. Ask the user for confirmation.
3. Open the phone dialler.

The app does **not** dispatch an ambulance itself.

Only verified official emergency numbers should be used.

2.5 Basic Medication Reminder

Users can create simple medication reminders.

Supported reminders can include:

- One-time reminders
- Recurring reminders
- Medication name
- Time
- Frequency

Notifications can be delivered through device push notifications, with SMS/WhatsApp considered later.

This feature is a **reminder tool**, not a prescribing or medical-advice system.

2.6 Community Exercise Planner

Users can create local community wellness activities.

Examples:

- Group walks
- Running groups
- Aerobics
- Stretching sessions
- Outdoor workouts
- Community wellness activities

An event can contain:

- Activity name
- Date and time
- Location
- Description
- Capacity
- RSVP option
- Reminders

Example:

Saturday Community Walk Saturday — 07:00 Local Community Park 18 people attending

This adds a **prevention and community-health dimension** to the platform rather than focusing exclusively on healthcare facilities.

2.7 Multilingual & Inclusive Access

The app should support **multiple South African languages** so that healthcare-access information is not restricted to English.

The interface should be designed so users can select their preferred language.

Potential supported languages include:

- English
- isiZulu
- isiXhosa
- Sesotho
- Sepedi
- Setswana
- siSwati
- Tshivenda
- Xitsonga
- Afrikaans
- isiNdebele

Language support should cover important user-facing areas such as:

- Navigation
- Facility information
- Queue reporting
- Forum categories
- Notifications
- Instructions
- Safety/privacy messages
- Basic feature descriptions

The implementation can begin with a smaller set of languages and expand over time.

For community-generated content, translation can be considered separately from interface localisation. User posts should retain their original language while optional translation can make information accessible to users who speak another language.

3. South African Health Data Resources

3.1 Facility & Service Information

Facility information can include:

- Location
- Opening hours
- Contact information
- Available services
- Emergency numbers
- Other relevant operational information

Some information can come from public health datasets, while current operational information can be crowdsourced.

4. Community Data & Trust

4.1 Location Verification

Users can optionally enable location when submitting a queue or facility report.

The system can verify that the user is physically near the selected facility.

Instead of storing continuous location history, the system can record only:

Location verified

along with the report timestamp.

Exact user locations should not be publicly displayed.

4.2 Data Confidence

Facility information can display a confidence level based on:

- Number of recent reports
- Report freshness
- Location verification
- Agreement between reports
- Historical reliability of contributors

For example:

High congestion — High confidence 9 recent reports · 6 location verified

This helps users distinguish between a single unconfirmed report and information supported by multiple independent users.

4.3 Community Reliability

Users can gradually build a reputation based on the usefulness and accuracy of their contributions.

Factors can include:

- Verified reports
- Reports confirmed by other users
- Useful forum contributions
- Reports that consistently agree with later community information

The system should reward **quality rather than quantity** to reduce spam.

5. AI / Machine Learning

AI should be used where it converts community-generated information into something more useful.

5.1 Content Classification

Forum posts can be automatically classified into categories such as:

- Queue report
 - Facility/service update
 - Medicine availability
 - Question
 - General discussion
-

5.2 Similarity & Report Aggregation

The system can identify posts describing the same situation.

For example:

"The queue is crazy today."

"I've been waiting forever."

"There are about 50 people here."

"Registration is extremely busy."

NLP similarity models can recognise that these posts are related and aggregate them into structured facility information.

This transforms:

Unstructured community reports → structured healthcare-access information

5.3 Duplicate & Spam Detection

AI can help identify:

- Duplicate reports
- Spam
- Suspicious activity
- Abusive content
- Contradictory information

Flagged content can be reviewed or moderated.

5.4 Future Predictive Analytics

Once enough historical data exists, the platform could eventually estimate:

- Expected congestion
- Expected waiting conditions
- Unusual facility activity

These should remain future features because meaningful predictions require sufficient historical data.

6. Privacy & Compliance

The app should minimise the personal information it collects.

Users should not be encouraged to submit:

- Patient names
- Patient medical records

- Patient photographs
- Confidential patient information
- Private staff information
- Passwords or internal access credentials

Facility staff can participate as ordinary community users without the platform needing to collect or verify their employment identity.

However, users remain responsible for ensuring that their contributions do not breach professional, employment, confidentiality or other legal obligations.

The platform should:

- Follow POPIA requirements where applicable.
- Collect only necessary information.
- Clearly explain location usage.
- Avoid exposing precise user locations.
- Provide reporting/moderation mechanisms.
- Clearly distinguish community reports from verified official information.

7. Prioritised Feature List

Priority 1 — MVP

These form the core product.

1. Facility Map & Information

- Nearby clinics, hospitals and pharmacies
- Facility pages
- Address and contact information
- Services
- Travel distance/time

2. Live Queue Status

- Community queue reports
- Queue size
- Estimated waiting time
- Timestamp
- Low/Moderate/High status
- Recent-report aggregation

3. Facility-Specific Forum

- Queue updates
- Service/operational updates
- Medicine availability

- Questions
- Community discussion
- Upvotes and confirmations

4. Emergency Call Button

- Verified facility emergency numbers
- One-tap dialler access
- Confirmation before calling

5. Basic Medication Reminder

- One-time reminders
- Recurring reminders
- Push notifications

6. Community Exercise Planner

- Create local activities
- Activity location
- Date/time
- RSVP
- Community participation

7. Multilingual Support

- Multiple South African languages
- Localised interface
- Translated feature instructions and important notifications
- Future support for translated community content

Priority 2 — Trust & Intelligence

- Location-verified reporting
- Data confidence indicators
- Community reliability scores
- Medicine Price Finder
- Simple points/rewards system
- AI content classification
- Duplicate/spam detection
- Community confirmation of reports
- Facility operational updates

Priority 3 — Advanced Intelligence

- Predictive wait-time estimates
- Real-time congestion estimation
- Anomaly detection
- Advanced NLP report aggregation
- Medicine availability intelligence
- More advanced community rewards
- Expanded language translation

Priority 4 — Future Ecosystem

- Community non-emergency transport
- Partnerships with licensed transport providers
- Advanced reward redemption
- Advert-based rewards
- Referral incentives
- Donations and community funding
- Other healthcare-access partnerships

Emergency transport coordination should only be considered if an appropriate legal, safety and operational model is established.

8. Product Direction

The app should remain focused on **healthcare access and community health**, rather than becoming a general-purpose health app.

The strongest product loop is:

Find a facility → see what is happening → contribute information → communicate with others → manage your health → participate in community wellness.

The primary differentiator is the ability to turn **real-time community contributions into useful healthcare-access intelligence**.

AI, location verification, confidence scoring and multilingual support can strengthen this system without changing its fundamental purpose.

The platform should remain:

- Community-driven
- Accessible
- Multilingual
- Privacy-conscious
- Location-aware when necessary
- Data-driven
- Focused on healthcare access rather than medical diagnosis

→ builds historical dataset

AI

→ moderates + structures community data

ML

→ learns congestion/waiting patterns

App

→ displays:

-  Facility
-  Current congestion
-  Waiting-time information
-  Community updates
-  Historical patterns

That gives you a much more interesting technical project: **a crowdsourced healthcare-access intelligence platform**, rather than simply a clinic queue app.